

Comprehensive Research Overview: IIT Bombay Computer Science & Engineering Faculty – Current and Recent Research Activities

Below is a unified, detailed list of specific, state-of-the-art research topics currently pursued by CSE faculty at IIT Bombay (2021–2025). Each item reflects fine-grained problem statements, methodologies, or systems that students may contribute to.

- Abir De: differentiable neural surrogates for graph algorithms (subgraph isomorphism, maximum common subgraph via ISONET/XMCS); data-efficient subset selection using combinatorial and neurally guided methods (Selcon, Gradmatch, Subselnet, Fhashnet); neural architectures for submodular/set-function learning (Flexsubnet); human-in-the-loop decision pipelines with feature-query optimization (Genex); capacity- and preference-constrained information diffusion models.
- Ajit A. Diwan: structural girth-constrained graph decompositions; disconnected 2-factors in bridgeless cubic planar graphs; P_3 -decompositions of directed graphs; explicit Δ -edge-coloring of divisor lattices; cycle-length modulo-k problems in connectivity.
- Ajit Rajwade: compressed sensing matrix design and error bounds for MRI, CT, cryo-EM; nonadaptive group testing under membership errors; neural-network-based image moderation via compressed-sensing features; tomography reconstruction with unknown projection angles.
- Akash Kumar: spectral graph theory for planted structure recovery; low-rank matrix completion; robust PCA; graph-based reinforcement learning (semi-bandit, dueling bandits, batched bandits); temporal-point-process recommenders.
- Akshay S. S.: timed, recursive, distributed system verification; automated functional synthesis; formal analysis of probabilistic/dynamical systems in systems biology and CPS.
- Arpit Agarwal: adaptive and batched submodular maximization; parallel approximate maximum flows; dynamic submodular cover with limited adaptivity; multi-pass streaming bandit memory-regret trade-offs.
- Ashwin Gumaste: NLP for image data acquisition; smart compressed imaging for reduced acquisition resources; compressed-sensing-driven image/video restoration

(denoising, demosaicing, deblurring, inpainting); graph signal processing for image compression.

- Bharat G. Adsul: formal methods in concurrency, logics and games; geometric complexity theory.
- Bhaskaran Raman: MAC protocol design for aerial audio networking; CSMA/CA scheduling with deadlines in 802.11ax; NFV chain orchestration on heterogeneous hardware; crowdsourced barometer-based traffic estimation; AI-assisted handwritten answer grading; vaccine-coercion survey quantification.
- Charu Agarwal: graph-edit-distance search using neural set divergence; step-by-step mechanistic LLM arithmetic reasoning; text-to-SQL compositional parsers with schema hallucination; Frugal LMs invoking symbolic solvers.
- Chetan Kamath: statically hiding batch proofs; PPAD vs. LWE hardness; verifiable delay functions from Lucas sequences; adaptive security limits of Yao's garbling; treewidth separators in garbling.
- Deepak Phatak: DBMS performance evaluation; distributed client-server benchmarking; autograding systems for programming assignments; system throughput modeling.
- Milind A. Sohoni: sustainable agriculture system-dynamics modeling; rural water-supply self-sustainability trade-offs (household connections vs. solar pumps); COVID-19 intervention impact via system dynamics.
- Mythili Vutukuru: peer-to-peer system anonymity analysis; P4 recursion-based deep packet inspection; IoT-scale network function multicore scalability (AppSteer); DDIOsim microarchitecture simulator for data-direct I/O; programmable dataplane 5G user-plane acceleration.
- Nutan Limaye: determinant-polynomial VP-completeness variants; $AC^0[+]$ circuit lower bounds; #SAT for constant-depth circuits with PTF gates; fixed-depth multilinear circuit depth-hierarchy.
- Parag Kumar Chaudhuri: physics-based character/garment simulation; remeshing-free graph FE fracture and ductile/brittle material modeling; hand-tracking with local-scale adaptation; deep interactive surface creation from 3D sketches.
- Purushottam Kulkarni: process-trace micro-configuration mining; PLD-based modular process modeling; dynamic process refactoring via edge contraction; business-process consistency taxonomy.

- Pushpak Bhattacharyya: sarcasm detection via cognitive eye-tracking; entity-relationship extraction in biomedical texts; low-resource MT/transliteration for related Indic languages; WordNet development for Indian languages.
- R. K. Shyamasundar: liveness verification under weak memory via Büchi automata; security-survivable distributed system modeling; formal methods tool-chain integration for cryptographic protocol analysis.
- Rohit Gurjar: isolation lemma derandomization via geometric methods; quasi-NC parallel algorithms for bipartite matching; algebraic pseudorandomness for derandomized parallel complexity.
- Rushikesh K. Joshi: unconstrained crossword puzzle automation; business-process evolution trace language; configuration mining from process execution logs; method refactoring by successive contraction.
- S. Sudarshan: probabilistic query optimization; keyword querying over structured/semi-structured data; query processing for noisy unstructured sources using graphical models.
- Sanjay A. Seshia: cyber-physical system verification; automated program synthesis; SMT and Horn-clause-based model checking; dependable computing architectures.
- Sayanddeep Saha: fault and side-channel cryptanalysis for hardware/EDA security; microarchitectural attack detection and mitigation.
- Sharath Chandran: virtual/augmented reality hand tracking; physics-based fluid and deformable solid animation with haptic feedback; OCR for scene text in Indic scripts; garment simulation.
- Shivaram Kalyan Krishnan: reinforcement learning for humanoid fall control; parameterized RL method characterization; predictive models for bipedal robot stabilization.
- Sruthi Sekar: dynamic geometric graph matching for subgraph retrieval; locality-sensitive hashing in Fourier domain for soft set containment; low-resource text+SQL multimodal question answering.
- Sujoy Bhore: dynamic maximum independent set of disks in $O(\text{polylog})$ update; fully dynamic MIS of disks; $O(1+\epsilon)$ approximation for geometric knapsack on spheres; NC algorithms for geometric bipartite matching.
- Sunita Sarawagi: sequence models for text/time-series; domain adaptation via human intervention; graphical/structured learning for data analytics; surprise-pattern mining in

OLAP cubes.

- Supratik Chakraborty: constrained model counting and sampling; counterexample-guided knowledge compilation; full-program induction for array programs without loop invariants; synthesis of Skolem functions.
- Umesh Bellur: autonomic cloud VM placement via interference-aware consolidation; distributed complex event processing; geo-spatial information system performance.
- Varsha Apte: queuing-theoretic overload control of web servers and WLANs; VM-performance interference modeling for datacenter applications; power-aware adaptive service orchestration.
- Vijay Ganesh: SAT/SMT solver development for AI and software-engineering; neurosymbolic integration of logic and neural methods; proof complexity of modern solvers.
- Vinay J. Ribeiro: secure consensus-throughput regulatory frameworks for cryptocurrencies; smart-contract computation maximization in PoW blockchains; martingale-based PAC mode estimation; packet-batching for botnet detection.